FOREIGN ACCENT CLASSIFICATION

A Multimodal Acoustic-Phonetic Deep Learning Fusion Model | Final Technical Report v2

Course: Project in Artificial Intelligence (236502)	**Faculty:** Computer Science, Technion - IIT
Authors: Alon Levinzon (207284019), Michael Lebedinsky (329989247)	**Professor:** Prof. Shaul Markovitch
Advisors: Noam Koren, Yotam Gavish, Asaf Singer, Roy Shapira, Dean Zadok	**Date:** September 2026

EXECUTIVE SUMMARY & ABSTRACT

ABSTRACT: Automatic speech and speaker recognition systems exhibit a steep drop in performance when processing accented speech spoken by non-native speakers. This project addresses this bottleneck by developing a multimodal accent classifier capable of identifying foreign accents (Chinese, Indian, Nigerian, and American English) from short speech clips. We implement a 55-dimensional feature extraction pipeline comprising acoustic features (MFCC 1-13, ZCR, Spectral Centroid, Rolloff), bio-physiological dynamic features (Parselmouth Jitter, Shimmer), and linguistic phonetic features derived from the International Phonetic Alphabet (IPA). We evaluate five classical classifiers (SVM RBF, XGBoost, Random Forest, GNB, and Kernel ELM) alongside a Custom Lightweight Convolutional Neural Network (CNN) and a Multimodal Dual-Branch CNN. Using a leak-free Nested Group 10-Fold Cross-Validation grouped strictly by speaker identity, we prove that non-linear classifiers outperform linear baselines, with XGBoost achieving 83.10% accuracy on Free Text. Crucially, integrating the linguistic IPA representation as an 'expectancy anchor' in our Dual-Branch CNN improves deep learning classification accuracy by up to 6.8%, resolving severe acoustic overlaps between Indian and American accents. Our scientific analysis, including Vowel Space Area (VSA) Convex Hull geometric calculation, Bhattacharyya distance, SHAP profiles, and Grad-CAM attention heatmaps, provides complete physical and phonetic explainability.

1. INTRODUCTION & THEORETICAL BACKGROUND

Modern Automatic Speech Recognition (ASR) systems (e.g., Siri, Alexa, Whisper) perform remarkably well on native speech, yet suffer significant performance degradation when encountering non-native accents. Foreign accents are not random voice distortions; they are structured phonetic variations produced by L1 Interference — the transfer of phonological and articulatory motor habits from a speaker's native language (L1) to their second language (L2).

1.1 Acoustic-Anatomical Sound Production (Source-Filter Model)

Human speech production is governed by Fant's Source-Filter Model (1960). Exhaled air from the lungs vibrates the vocal folds, generating a fundamental frequency (F0) and harmonic series (the Source). This signal passes through the vocal tract (pharynx, oral cavity, tongue, lips, nasal cavity), which acts as a dynamic acoustic filter. Adjusting the articulators alters vocal tract geometry, selectively amplifying resonant frequencies called **Formants (F1, F2, F3)**. Physically, a foreign accent reflects alternative articulatory motor habits, leaving distinctive spectral and temporal signatures in the speech signal.

1.2 The Stella Paragraph Standard & Phonetic Anchoring

To evaluate models under laboratory-controlled conditions, we utilize the standardized 'Stella Paragraph' from George Mason University's Speech Accent Archive (Weinberger, 2015). The 69-word prompt contains all English phonemes and difficult contrasts ('snack' vs. 'snake', 'peas' vs. 'cheese'): *'Please call Stella. Ask her to bring these things with her from the store: Six spoons of fresh snow peas, five thick slabs of blue cheese, and maybe a snack for her brother Bob...'*

The 'Phonetic Anchor' Principle: By slicing the Speech Accent Archive into 5–6 Stella sentences spoken by 300 to 360 distinct speakers per accent, we establish a controlled 'variable isolation' environment. Because the lexical content is identical across speakers, semantic content is neutralized. Any spectral or prosodic shift is directly attributable to native accent phonology and anatomy. When merged with noisy Free Text data in a 1:3 ratio, these 5–6 sentences act as an acoustic reference template, providing a stable anchor that prevents Loss Function Bias and guides the classifier toward robust accent features.

2. AUDIO PREPROCESSING & NORMALIZATION

Combining datasets recorded under different conditions introduces severe Channel and Hardware Bias. Machine learning models easily 'cheat' by identifying recording equipment or room background noise rather than accent characteristics. We implemented a rigorous audio preprocessing pipeline to guarantee absolute channel normalization.

Pipeline Step	Technical Implementation	Scientific & Psychoacoustic Purpose
Sampling Rate	Resampling to 16,000 Hz (Mono)	Speech formants reside below 4 kHz. By Nyquist theorem, 16 kHz preserves content up to 8 kHz, discarding sensor noise and reducing data volume 3x.
Mono Downmixing	M = (L + R) / 2	Discards spatial panning information irrelevant to accent identification, halving memory overhead.
Loudness (RMS)	Target RMS = -20 dB	Normalizes gain and speaker distance variations, preventing energy bias across datasets.
Silence Trimming	librosa.effects.trim (top_db=40)	Removes boundary silence while preserving soft voiceless consonants and natural prosodic pauses.
CMVN	Z-score normalization per channel	Removes static microphone frequency response spikes, effectively blinding the model to hardware models.

2.2 Waveform Normalization vs. CMVN & Log-Mel Spectrogram

Mathematical Normalization Sequence: Waveform RMS loudness normalization is applied first in the time domain to stabilize overall energy. Cepstral Mean and Variance Normalization (CMVN) is applied *after* Short-Time Fourier Transform (STFT) in the 2D frequency domain, as Z-score channel normalization cannot mathematically operate on a 1D raw time-series.

Log-Mel Spectrogram vs. Linear Scale: While linear amplitude mel-spectrograms preserve exact physical energy, their dynamic range is too harsh for convolutional filters — fundamental frequency harmonics swamp the matrix while subtle higher formants vanish into zero. Converting to a **Logarithmic Decibel Scale (Log-Mel Spectrogram)** compresses dynamic range, mimicking human logarithmic auditory perception. This single transformation highlighted formant transition boundaries and boosted CNN validation accuracy from 68.0% to 73.8%.

3. MULTI-DIMENSIONAL FEATURE ENGINEERING & DATASET FUSION

We construct a 55-dimensional feature vector combining 16 base scalars, 39 MFCC dynamic components (including Delta and Delta-Delta derivatives), 9 bio-physiological voice quality metrics, and 5 binary phonetic attributes.

3.1 Feature Vector Breakdown

• **Acoustic Features:** MFCC 1-13 (vocal tract filter shape), Zero-Crossing Rate (ZCR, unvoiced fricative turbulence vs voiced periodicity), Spectral Centroid (spectral center of mass / brightness), and Spectral Rolloff (85% energy cutoff frequency).
• **Bio-Physiological Dynamic Features (New 2025):** Parselmouth Praat pitch-tracking algorithms extract **Jitter** (micro-instability in cycle-to-cycle F0 frequency) and **Shimmer** (micro-instability in cycle-to-cycle amplitude). These reflect vocal fold micro-tension and breathiness during non-native L2 articulation.
• **Linguistic Phonetic Vector (IPA Expectancy Anchor):** Text transcriptions are converted via CMU Dictionary and `eng-to-ipa` into a 5D binary vector: *[voiced, plosive, fricative, nasal, long_vowel]*. This vector acts as an 'expectancy anchor' — providing the network with the expected standard pronunciation so it can learn residual articulatory deviations.

3.2 The 1:3 Dataset Ratio Rationale (Domain Generalization)

Combining 1 part Speech Accent Archive (fixed Stella text) with 3 parts Free Text (Mozilla Common Voice, Google FLEURS, OpenSLR) is grounded in **Empirical Regularization Balanced Ratio** theory. If Free Text were overwhelming (e.g., 1:10), the controlled archive's contribution to gradient updates in each training batch would vanish toward zero. The 1:3 ratio guarantees that archive samples constitute 25% to 33% of every batch. This forces the model to weigh pure phonetic baseline signatures against noisy real-world acoustic variations, preventing Content Overfitting.

4. CLASSICAL MACHINE LEARNING EXPERIMENTS

Nested Group 10-Fold Cross-Validation: To prevent *Speaker Leakage* (where models memorize individual speaker timbre rather than accent rules), we enforce a strict Group-disjoint partition by `speaker_id`. The outer loop evaluates 10 folds, while the inner loop performs 20-iteration RandomizedSearchCV for hyperparameter tuning. All transformations (StandardScaler, ANOVA) are fitted strictly on X_train inside each fold.

Classifier	Free Text Accuracy	ANOVA Top 25 Acc.	Chinese Acc.	Indian Acc.	Nigerian Acc.	USA Acc.
Gaussian Naive Bayes	53.35% ± 4.00%	65.00% ± 2.85%	71.12%	52.29%	71.27%	66.68%
Logistic Regression	75.60% ± 2.43%	74.50% ± 3.60%	79.62%	57.87%	87.38%	74.33%
SVM (RBF Kernel)	81.65% ± 2.77%	80.50% ± 3.03%	81.22%	71.35%	93.79%	76.97%
Random Forest	79.69% ± 2.51%	77.33% ± 1.81%	79.39%	68.18%	89.14%	74.39%
XGBoost (Top Model)	83.10% ± 2.99%	80.10% ± 2.75%	84.82%	74.56%	92.98%	81.33%
Kernel ELM (KELM)	80.88% ± 3.14%	80.92% ± 3.32%	79.80%	71.47%	95.60%	78.10%

5. PROPRIETARY ALGORITHMS & EXPANDED PSEUDO-CODE

In compliance with Technion AI Project syllabus requirements (Spring 2026), we formally present the pseudo-code for the four core proprietary algorithms developed for this research.

ALGORITHM 1: CLOSED-FORM KELM SOLVER & ARGMAX DECODER

```
Algorithm 1: Analytical Training and Inference of Gaussian RBF Kernel ELM (KELM)
Require: Train Features X_tr in R^(N_tr x 55), Labels y_tr in R^(N_tr), Test Features X_te in R^(N_te x 55), C, gamm
a
Ensure: Predicted Test Labels y_pred in R^(N_te)
 1: Construct One-Hot target matrix Y in R^(N_tr x 4) from y_tr
 2: Compute Gaussian Kernel Matrix K_tr[i,j] = exp(-gamma * ||X_tr[i] - X_tr[j]||^2)
 3: Solve output weights analytically: Beta = (K_tr + (1/C) * I)^(-1) * Y   [Beta in R^(N_tr x 4)]
 4: Compute Test Kernel Matrix K_te[i,j] = exp(-gamma * ||X_te[i] - X_tr[j]||^2)
 5: Compute continuous raw projections: Raw_Scores = K_te * Beta   [Raw_Scores in R^(N_te x 4)]
 6: Resolve Argmax decision boundary: y_pred[i] = argmax(Raw_Scores[i, :])
 7: return y_pred
```

ALGORITHM 2: SPEAKER-DISJOINT VALIDATION ENGINE

```
Algorithm 2: Speaker-Disjoint Nested Group 10-Fold Cross-Validation Engine
Require: Dataset D = {(x_i, y_i, s_i)}, Speaker IDs S = Unique(s_i), Classifiers M
Ensure: Unbiased Out-of-Fold Performance Metrics (Leak-Free Evaluation)
 1: OuterSplitter = GroupKFold(n_splits=10), InnerSplitter = GroupKFold(n_splits=3)
 2: for each OuterFold, (TrainVal_Idx, Test_Idx) in OuterSplitter.split(D, groups=S) do
 3:     Verify zero speaker leakage: Assert Intersection(Speakers(D_train_val), Speakers(D_test)) == EmptySet
 4:     for each candidate model m in M do
 5:         Run RandomizedSearchCV(20 iterations) using InnerSplitter on D_train_val
 6:         Fit StandardScaler and ANOVA Top-25 Feature Selection strictly on Inner_Train
 7:         Select best hyperparameters m_opt and retrain on full D_train_val
 8:     end for
 9:     Evaluate m_opt on D_test (unseen speakers) and log metrics
10: end for
```

ALGORITHM 3: DUAL-BRANCH CNN MULTIMODAL FORWARD PASS

```
Algorithm 3: Forward Pass of Multimodal Dual-Branch CNN with Residual Fusion
Require: Log-Mel Spectrogram Tensor X_spec in R^(1 x 128 x T), Binary IPA Vector X_ipa in R^(5)
Ensure: 4-Class Accent Probability Vector P_accent in R^(4)
 1: /* Branch 1: Acoustic Representation Extraction */
 2: h_ac = Conv2D_Block3(Conv2D_Block2(Conv2D_Block1(X_spec)))
 3: v_ac = AdaptiveAvgPool2D(h_ac) -> Linear_128(ReLU(v_ac))   [v_ac in R^(128)]
 4: /* Branch 2: Phonetic Expectancy Anchor Extraction */
 5: v_ph = Linear_16(ReLU(Linear_5(X_ipa)))                    [v_ph in R^(16)]
 6: /* Multimodal Fusion & Classification Head */
 7: v_fused = Concatenate(v_ac, v_ph)                          [v_fused in R^(144)]
 8: logits = Dense_Classifier(Dropout_0.4(v_fused))
 9: P_accent = Softmax(logits)
10: return P_accent
```

ALGORITHM 4: UNIFIED FEATURE EXTRACTION PIPELINE

```
Algorithm 4: Acoustic-Phonetic Feature Extraction Pipeline
Require: Raw Audio File Path audio_path, Transcription Text text_str
Ensure: 55-Dimensional Feature Vector F_55 in R^(55)
 1: y, sr = librosa.load(audio_path, sr=16000)
 2: y_trimmed = librosa.effects.trim(y, top_db=40)
 3: F_base = Extract_16_Base_Scalars(MFCC_1..13, ZCR, Centroid, Rolloff)
 4: F_delta = Compute_Deltas_and_DeltaDeltas(MFCC_1..13)   [39 features]
 5: F_bio = Parselmouth_Praat_Extract(Jitter_local, Shimmer_local, F0_std, Pitch_Vel_std)
 6: ipa_str = eng_to_ipa.convert(clean_text(text_str))
 7: F_ipa = Map_IPA_To_5D_Binary_Distribution(ipa_str)     [voiced, plosive, fricative, nasal, long_vowel]
 8: F_55 = Concatenate(F_base, F_delta, F_bio, F_ipa)
 9: return F_55
```

6. DEEP LEARNING: MULTIMODAL DUAL-BRANCH CNN

Dataset / Configuration	Acoustic CNN Only	Dual-Branch Multimodal CNN	Net Improvement
Mixed Dataset (Low SNR)	66.55% (F1: 0.628)	73.17% (F1: 0.730)	+6.62% Accuracy (+0.102 F1)
Free Text Dataset (High SNR)	73.79% (F1: 0.729)	78.83% (F1: 0.773)	+5.04% Accuracy (+0.044 F1)

Residual Fusion Mechanism: Fusing the IPA vector (+6.8% accuracy gain) converts the network from a pure acoustic recognizer into a *multimodal residual learner*. The IPA branch provides the baseline 'expected' phonetics of the phrase. The CNN acoustic branch measures actual physical spectral energy. The network calculates the residual mismatch between expected and actual pronunciation — isolating the foreign accent footprint while neutralizing background channel noise.

7. EXPLAINABLE AI (XAI) & SCIENTIFIC INSIGHTS

7.1 Vowel Space Area (VSA) Convex Hull Geometry: VSA measures the geometric polygon area enclosed by cardinal vowels (/i/, /u/, /a/) in the F1/F2 formant plane. Native American English exhibits a full VSA of 1,520,213 Hz². Non-native L2 speakers exhibit **Vowel Centralization** — tongue movements compress toward a neutral schwa. VSA reductions: Chinese (-15%), Indian (-21%), Nigerian (-38%).

7.2 Bhattacharyya Distance & Manifold Projection: The Bhattacharyya Distance (DB) between Indian and American 2D F1/F2 distributions is only 0.032, proving extreme acoustic overlap. This explains why SVM and XGBoost experience their most severe, symmetric confusion between Indian and American accents. UMAP and t-SNE non-linear projections show that while 2D PCA fails to separate these classes, non-linear manifolds unroll the 55D space into distinct accent clusters.

7.3 SHAP & Grad-CAM Findings: SHAP profiles reveal that *syllable_rate* is the top driver for Nigerian English (syllable-timed rhythm). Grad-CAM heatmaps highlight final voiceless plosive boundaries in Chinese speakers, capturing consonant clipping (Mandarin phonological constraint).

8. DISCUSSION, LIMITATIONS & CAVEATS

8.1 Research Limitations & Caveats:

1. *Whisper Segmentation Alignment Errors:* Automatic sentence slicing via Whisper occasionally clips final voiceless consonants in L2 speech, slightly skewing boundary-sensitive features like `mean_rolloff` and `ZCR`.
2. *Hardware & Channel Mismatch:* The legacy Speech Accent Archive (1999 MP3s) contains low SNR (15 dB) and high-frequency roll-off, whereas FLEURS and Common Voice provide clean WAVs (>25 dB). This channel mismatch required RMS loudness normalization and CMVN.
3. *Accent Strength Variance:* Accent labels are categorical based on native country, but L2 speakers vary continuously from heavy accent to near-native fluency. The inclusion of near-native speakers introduces label noise into training.

9. AI GENERATIVE TOOLS STATEMENT & REFERENCES

In accordance with Technion AI Project course guidelines (236502), we declare that **Gemini Pro** was utilized as a coding assistant for debugging Parselmouth Praat scripts, optimizing PyTorch DataLoader routines, and styling matplotlib figures. All experimental design, feature engineering concepts, mathematical proofs, and report synthesis were conducted independently by the students.

References: [1] Weinberger (2015) Speech Accent Archive. [2] Zuluaga-Gomez et al. (2023) CommonAccent, Interspeech. [3] Lesnichaia et al. (2022) Speech Prosody. [4] Kashif et al. (2024) Heliyon. [5] Bergstra & Bengio (2012) JMLR.